Session 29: Consolidated Financial Statements: The Concept of Control II (Intra-Group Eliminations and NCI)

By the end of this session you must know how to eliminate intra-group balances and unrealised profits (on inventory and on fixed assets), how post-acquisition consolidation adjustments (depreciation of step-ups, deferred tax reversals) flow through, how intra-group dividends are eliminated, and how to compute non-controlling interests' share of net income.

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1. Intra-Group Balances

Intra-group balances are cancelled out entirely when preparing the group statement of financial position. They arise from: one group company lending to another (a loan asset for the lender, a liability for the borrower). One group company selling goods/services on credit to another (a receivable for the seller, a payable for the buyer), or a current account balance between group companies (an asset for one, a liability for the other).

Intra-group loan: Alfa lends €10,000 to Beta, and Beta pays €500 interest to Alfa during the year. On consolidation, Alfa's loan receivable (10,000) is cancelled against Beta's loan payable (10,000), and Alfa's interest income (500) is cancelled against Beta's interest expense (500). Both sides of the internal transaction disappear entirely from the group accounts, since a group cannot owe money to, or earn interest from, itself.

2. Unrealised Profits: The General Rule

Unrealised profit: profit embedded in an asset's carrying value because it was sold from one group company to another at a price above the seller's original cost, where the asset has not yet been sold outside the group.

Must know (elimination rule, IFRS10):
• Profits/losses on intra-group transactions are eliminated **in full** on consolidation, regardless of whether the subsidiary involved is wholly- or partly-owned
• **Parent sells to subsidiary (downstream):** unrealised profit is subtracted entirely from **group retained earnings** (none of it is charged to NCI, since the parent, not the partly-owned subsidiary, generated the profit)
• **Subsidiary sells to parent (upstream):** unrealised profit is allocated **proportionately** between group retained earnings and NCI (since it was the partly-owned subsidiary that generated the profit)

3. Consolidation Adjustments: Depreciating Step-Ups and Reversing Deferred Tax

- Unrealised inventory profit reduces the group's reported inventory (and correspondingly increases cost of sales in the group statement of comprehensive income, since a lower closing inventory pushes more cost through to COGS).
- Goodwill, as always, is never amortised. Any impairment loss is expensed directly (IAS36, Sessions 12-13).
- Fair-value step-up/step-down adjustments recognised at acquisition (Session 28) must then be carried forward and followed through post-acquisition: if linked to a depreciable/amortisable asset, the step-up itself must be depreciated/amortised over the asset's remaining useful life. If the related asset is no longer present at the consolidation date (e.g. it was sold), the adjustment must be reversed through profit or loss.
- Deferred tax effects must be recognised for all of these adjustments.

Continuing the Session 28 Alfa/Beta example (PP&E step-up €1,500 over 20 years remaining life, Patents step-up €1,000 over 10 years remaining life, tax rate 50%): the incremental depreciation on the PP&E step-up = 1,500 ÷ 20 = €75/year. The incremental amortisation on the patents step-up = 1,000 ÷ 10 = €100/year. Total additional depreciation/amortisation charged in the consolidated income statement = 75 + 100 = €175. Because this charge reduces consolidated profit, part of the deferred tax liability originally recognised on the step-ups (Session 28) must be reversed: reversal = 50% × 175 = **€87.50**. Consolidated entries: Dr Depreciation (PPE) 75, Dr Amortisation (Patents) 100 [reducing PPE and Patents' carrying amounts by the same totals], and Dr DT Liability 87.50, Cr Deferred Income Tax Expense 87.50 (a tax benefit, partially offsetting the extra depreciation/amortisation expense).

4. Intra-Group Items in the Income Statement

Group statement of comprehensive income: reports the profit or loss of the group as a single economic entity, prepared by adding together (line by line) every group company's individual statement, then cancelling out intra-group items. The income-statement mirror of the balance-sheet consolidation process.

Must know (items to cancel in the group income statement):
• **Intra-group sales:** deducted from both group sales revenue **and** group cost of sales (the two sides of the same internal transaction)
• **Intra-group interest, management fees** and similar intra-group charges: cancelled out
• **Dividends paid by a subsidiary to the parent:** cancelled against the dividends the parent recorded as income (Section 8 below)
• All these eliminations are made **in full**, even where NCI is involved
• If a consolidation adjustment creates a temporary difference affecting the consolidated income statement, the related deferred tax effect must be recognised (exactly as in Section 3)

5. Unrealised Profit on Inventory: A Full Worked Example

Setup: Beta sells goods to Alfa for €5,000. Beta had originally bought them for €4,000 (intra-group profit €1,000). At year-end, all the goods remain unsold in Alfa's inventory.

Elimination: from the group's perspective, the entire €1,000 profit is unrealised (the goods have not left the group) and must be eliminated: consolidated sales are reduced by 5,000, consolidated COGS is reduced by 4,000 (net effect: consolidated profit falls by 1,000), and consolidated inventory is written down by 1,000 (from the 5,000 transfer price back to the original 4,000 cost). Because Beta already recognised income tax on this profit, but the profit is not yet realised from the group's perspective, a **deferred tax asset** is recognised: DTA = 50% × 1,000 = **€500**.

Symmetrically, if the intra-group transaction had instead generated a *loss* (a negative margin), the inventory would need to be written *back up* in the consolidated accounts, with a corresponding deferred tax **liability** recognised instead.

6. In-Depth: Inventory Write-Downs and Genuine Losses

- This section addresses a subtlety: what if the buying company has *already* partially written the inventory down to a lower fair value by year-end, for reasons unrelated to the intra-group markup?

Alfa sells goods to Beta for €1,200 (originally cost Alfa €1,000, intra-group profit €200). All goods remain in Beta's year-end inventory, held at the lower of cost and fair value.

(A) Fair value = €1,150: Beta already wrote inventory down from 1,200 to 1,150 (a €50 write-down). Since €50 of the original €200 intra-group profit has already been implicitly reversed via that write-down, only the remaining **€150** is still unrealised and must be eliminated in consolidation.
(B) Fair value = €1,000: Beta wrote inventory down to 1,000 (a €200 write-down), exactly absorbing the entire €200 intra-group profit already. **No further adjustment** is needed.
(C) Fair value = €900: Beta wrote inventory down to 900 (a €300 write-down), more than absorbing the €200 profit. The intra-group profit is again fully eliminated with **no further adjustment** required (the excess write-down below original cost reflects a genuine decline in the asset's value, not a consolidation issue).

Genuine losses are not reversed: if a parent sells to a subsidiary *below* cost because the sale price matches the goods' actual fair value at that time, no consolidation reinstatement is needed. The seller would have had to write the inventory down to that same fair value even without the intra-group sale, so there is no artificial "loss" created by the internal transaction to eliminate.

7. Unrealised Profit on a Fixed Asset Sale

Setup: Alfa sells a fixed asset to Beta for €500. The asset's net book value on Alfa's books was €200 (intra-group gain €300). By year-end, Beta has charged €100 of depreciation on the asset (based on its new €500 cost). Had the asset stayed with Alfa, depreciation would only have been €80 (based on the original €200 cost). Absent the sale, the group's net book value for the asset would simply be 200 − 80 = **€120**.

Three-part elimination: (1) eliminate the €300 gain on disposal Alfa recognised, (2) reduce depreciation by €20 (Beta recorded 100, but the group-level figure, based on the original €200 cost, would only be 80), (3) adjust the asset's carrying amount by reversing €280, bringing it from Beta's post-sale net book value of €400 (500 cost − 100 depreciation) down to the group-level €120. Since these three adjustments together reduce consolidated profit by 300 − 20 = €280 net, a deferred tax asset is recognised: DTA = 50% × 280 = **€140**.

8. Intra-Group Dividends

Intra-group dividend: effectively an intra-group resource (equity) transfer. A subsidiary distributing part of its own retained earnings to its parent (and, proportionately, to any NCI). A consolidation adjustment removes the distribution from the group accounts, since the group cannot pay a dividend to itself.

Beta (80%-owned by Alfa) distributes total dividends of €1,000. Alfa receives its 80% share (800) as dividend income. The NCI shareholders receive their 20% share (200) directly from Beta, outside the group. On consolidation: the 800 Alfa recorded as dividend income is eliminated (cancelled against the corresponding reduction in Beta's retained earnings). The remaining 200 paid to non-controlling shareholders reduces NCI's equity stake by that same amount (since it is a genuine cash outflow to parties outside the group). Net effect: the 800 intra-group dividend is eliminated against the corresponding distribution. The 200 paid outside the group reduces NCI equity. The group's retained earnings attributable to the parent do not fall by the full 1,000 dividend.

9. Attributing Net Income to Non-Controlling Interests

Must know (the golden rule of consolidation):
• Every subsidiary's assets, liabilities, revenues and expenses are consolidated at **100%**, regardless of the parent's actual ownership percentage. And every consolidation adjustment (step-up depreciation, unrealised profit elimination, etc.) is likewise applied at 100% first
• Only **after** full (100%) consolidation is the minority shareholders' share separated out, as: **NCI in equity** (determined after eliminating the investment, adjusted for items like intra-group dividend eliminations) and **NCI in net income** (computed at the very end of the consolidation process)

NCI's share of net income = % NCI × (Subsidiary's net income ± consolidation adjustments affecting the subsidiary's net income)

where: - "consolidation adjustments affecting the subsidiary's net income" means adjustments originating in that specific subsidiary (e.g. step-up depreciation on assets it holds, unrealised profit it generated on an upstream sale). Not adjustments originating with the parent (downstream sales), which per Section 2 are charged entirely to group retained earnings, never to NCI.

Continuing the Alfa/Beta example: Beta's net income for the year = €1,000. NCI = 20%. Consolidation adjustments affecting Beta's own profit: (1) depreciation of step-ups, net of tax = 175 gross − 87.5 tax reversal = €87.50 net reduction to income, (2) intra-group unrealised inventory profit, net of the deferred tax asset = 1,000 gross − 500 DTA = €500 net reduction to income. NCI's share of net income = 20% × (1,000 − 87.5 − 500) = 20% × 412.5 = **€82.50**.

Recognition: Dr Net income, Group 82.50. Cr Net income, NCI 82.50 (in the consolidated income statement, reallocating that portion of profit away from the group's own shareholders), and correspondingly, Dr Equity, Group 82.50. Cr Equity, NCI 82.50 (in the consolidated balance sheet).

10. Common Mistakes and Exam Traps

- Allocating unrealised profit on a *downstream* sale (parent→subsidiary) partly to NCI. Downstream unrealised profit is entirely a group-retained-earnings adjustment, since it was the parent (100% group-owned) that generated it.
- Allocating unrealised profit on an *upstream* sale (subsidiary→parent) entirely to group retained earnings. Upstream profit was generated by the (partly-owned) subsidiary and must be split proportionately between group retained earnings and NCI.
- Forgetting that consolidation adjustments (step-up depreciation, unrealised profit elimination) must be applied at 100% before computing NCI's share, not net of NCI's percentage from the start.
- Reversing an intra-group loss that actually reflects a genuine decline in the asset's fair value. Only losses that are artefacts of internal pricing (not real value declines) get reinstated.
- Missing that a partially-absorbed inventory write-down (as in Example A, fair value 1,150) still requires a *partial* unrealised-profit elimination for the remaining, not-yet-absorbed portion.
- Forgetting the deferred tax consequence of every consolidation adjustment that hits the income statement. Unrealised profit eliminations and step-up depreciation reversals both require a matching DTA/DTL entry.

11. Must Know Absolutely

Must know:
• Intra-group balances (loans, trade receivables/payables, current accounts) are fully cancelled on consolidation
• Unrealised profit is eliminated in full. Downstream (parent→sub) hits group retained earnings only. Upstream (sub→parent) is split between group retained earnings and NCI
• Step-up/step-down adjustments continue post-acquisition: depreciate/amortise over remaining useful life (with a partial DTL/DTA reversal each period), or reverse entirely if the related asset leaves the group
• Inventory already written down for unrelated reasons absorbs some/all of the intra-group profit automatically. Only eliminate what remains unabsorbed. Genuine fair-value declines are never reversed
• Fixed-asset intra-group sale: eliminate the gain, reduce depreciation to the group's original-cost basis, adjust the carrying amount back to group-level NBV, recognise a DTA/DTL on the net P&L effect
• Intra-group dividends: eliminate the parent's dividend income against the subsidiary's distribution. Only dividends actually paid to NCI shareholders reduce NCI equity
• NCI's share of net income = %NCI × (subsidiary's net income ± adjustments originating with that subsidiary), computed only after 100% consolidation